

Term 3 — Lesson 1 Practice Activity (C4) with Solutions

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Evaluated criteria

- C4: Applies the general procedure for hypothesis testing with fixed probability in contextual situations.

1 General Problem (One-Sample Proportion Test Template)

A benchmark claim concerns a population proportion. A random sample of size n has x successes, so $\hat{p} = x/n$. At significance level α , test whether the true proportion is lower than, higher than, or different from the benchmark p_0 .

Reference table for common significance levels

Significance level	Upper-tail p-value	z_α such that $P(Z > z_\alpha) = \alpha$
10%	0.10	0.842
5%	0.05	1.645
2%	0.02	2.054
1%	0.01	2.326

Questions/Tasks.

- Identify the population parameter and write hypotheses for lower-tail, higher-tail, and two-tailed alternatives.
- State the condition checks for a one-sample proportion z -test.
- Write the one-sample proportion test statistic.
- State how the p-value is computed for lower-tail, higher-tail, and two-tailed tests.
- State the decision rule and write a contextual conclusion template.

C4

- **Parameter:** p = true population proportion for the context.

- **Hypotheses:**

$$H_0: p = p_0, \quad H_a: p < p_0 \text{ or } p > p_0 \text{ or } p \neq p_0.$$

- **Conditions:** Check $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

- **Test statistic:**

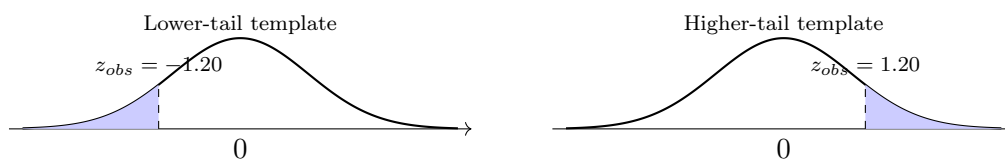
$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}.$$

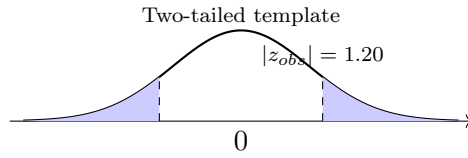
- **p-value by tail type:**

- Lower-tail: $P(Z \leq z_{obs})$
- Higher-tail: $P(Z \geq z_{obs})$
- Two-tailed: $2P(Z \geq |z_{obs}|)$

- **Decision rule:** Reject H_0 if p-value $\leq \alpha$; otherwise fail to reject H_0 .

- **Conclusion template:** “There is sufficient evidence” (reject H_0) or “There is not enough evidence” (fail to reject H_0) for the contextual claim in H_a .





2 On-time assignment completion rate (introductory higher-tail)

A school reports that 72% of assignments are submitted on time. Consider two random samples:

- Sample A has 250 assignments, of which 190 are submitted on time.
- Sample B has 500 assignments, of which 380 are submitted on time.

Notice that both samples have the same sample proportion. At $\alpha = 5\%$, test whether the true on-time proportion is higher than 72% for each sample.

C4

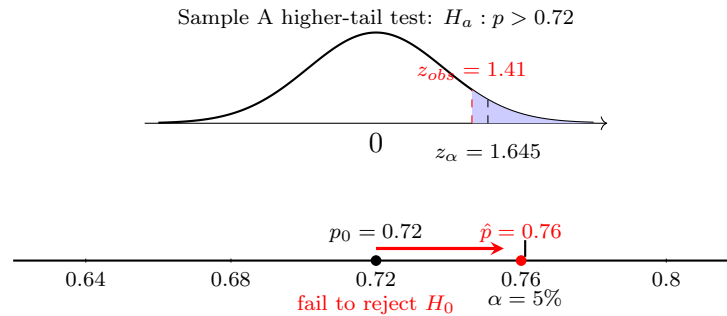
- **Parameter:** p = true on-time submission proportion.
- **Hypotheses:** $H_0: p = 0.72$, $H_a: p > 0.72$.

Sample A

- **Conditions:** $np_0 = 250(0.72) = 180 \geq 10$, $n(1 - p_0) = 250(0.28) = 70 \geq 10$.
- **Sample proportion:** $\hat{p} = 190/250 = 0.76$.
- **Test statistic:**

$$z = \frac{0.76 - 0.72}{\sqrt{\frac{0.72(0.28)}{250}}} \approx 1.41.$$

- **p-value:** $P(Z \geq 1.41) \approx 0.0795$.
- **Decision:** Fail to reject H_0 since $0.0795 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is above 72%.

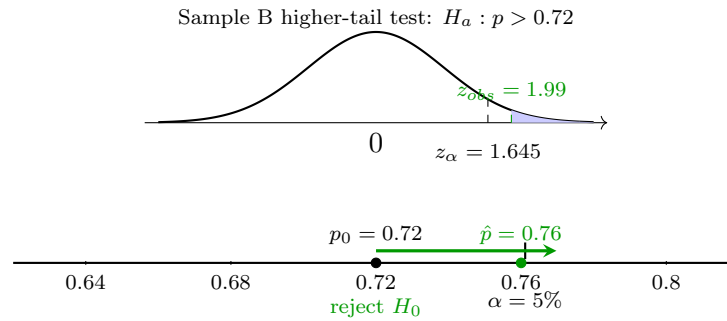


Sample B

- **Conditions:** $np_0 = 500(0.72) = 360 \geq 10$, $n(1 - p_0) = 500(0.28) = 140 \geq 10$.
- **Sample proportion:** $\hat{p} = 380/500 = 0.76$.
- **Test statistic:**

$$z = \frac{0.76 - 0.72}{\sqrt{\frac{0.72(0.28)}{500}}} \approx 1.99.$$

- **p-value:** $P(Z \geq 1.99) \approx 0.0232$.
- **Decision:** Reject H_0 since $0.0232 < 0.05$.
- **Conclusion:** There is enough evidence that the true on-time submission rate is above 72%.



Although both samples have the same sample proportion $\hat{p} = 0.76$, the larger sample gives a more positive test statistic and a smaller p-value because the standard error is smaller.

3 On-time assignment completion rate (introductory lower-tail)

A school reports that 72% of assignments are submitted on time. Consider two random samples:

- Sample A has 250 assignments, of which 170 are submitted on time.
- Sample B has 500 assignments, of which 340 are submitted on time.

Notice that both samples have the same sample proportion. At $\alpha = 5\%$, test whether the true on-time proportion is lower than 72% for each sample.

C4

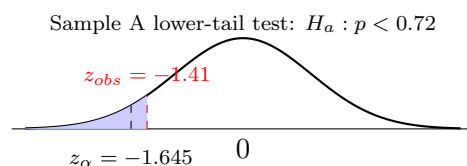
- **Parameter:** p = true on-time submission proportion.
- **Hypotheses:** $H_0 : p = 0.72$, $H_a : p < 0.72$.

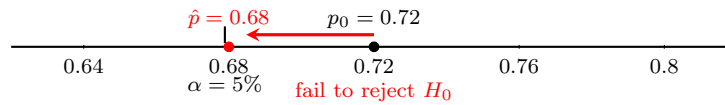
Sample A

- **Conditions:** $np_0 = 250(0.72) = 180 \geq 10$, $n(1 - p_0) = 250(0.28) = 70 \geq 10$.
- **Sample proportion:** $\hat{p} = 170/250 = 0.68$.
- **Test statistic:**

$$z = \frac{0.68 - 0.72}{\sqrt{\frac{0.72(0.28)}{250}}} \approx -1.41.$$

- **p-value:** $P(Z \leq -1.41) \approx 0.0795$.
- **Decision:** Fail to reject H_0 since $0.0795 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is below 72%.



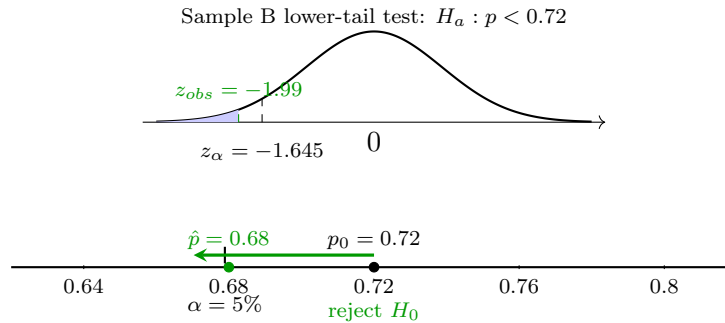


Sample B

- **Conditions:** $np_0 = 500(0.72) = 360 \geq 10$, $n(1 - p_0) = 500(0.28) = 140 \geq 10$.
- **Sample proportion:** $\hat{p} = 340/500 = 0.68$.
- **Test statistic:**

$$z = \frac{0.68 - 0.72}{\sqrt{\frac{0.72(0.28)}{500}}} \approx -1.99.$$

- **p-value:** $P(Z \leq -1.99) \approx 0.0232$.
- **Decision:** Reject H_0 since $0.0232 < 0.05$.
- **Conclusion:** There is enough evidence that the true on-time submission rate is below 72%.



Although both samples have the same sample proportion $\hat{p} = 0.68$, the larger sample gives a more negative test statistic and a smaller p-value because the standard error is smaller.

4 On-time assignment completion rate (introductory two-tailed)

A school reports that 72% of assignments are submitted on time. Consider two random samples:

- Sample A has 250 assignments, of which 190 are submitted on time.
- Sample B has 500 assignments, of which 380 are submitted on time.

Notice that both samples have the same sample proportion. At $\alpha = 5\%$, test whether the true on-time proportion is different from 72% for each sample.

C4

- **Parameter:** p = true on-time submission proportion.
- **Hypotheses at $\alpha = 5\%$:** $H_0: p = 0.72$, $H_a: p \neq 0.72$.

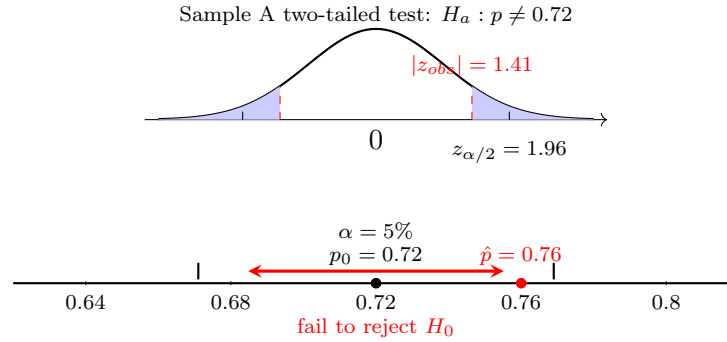
Sample A

- **Conditions:** $np_0 = 250(0.72) = 180 \geq 10$, $n(1 - p_0) = 250(0.28) = 70 \geq 10$.
- **Sample proportion:** $\hat{p} = 190/250 = 0.76$.

- **Test statistic:**

$$z = \frac{0.76 - 0.72}{\sqrt{\frac{0.72(0.28)}{250}}} \approx 1.41.$$

- **p-value:** $2P(Z \geq 1.41) \approx 0.1590$.
- **Decision:** Fail to reject H_0 since $0.1590 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is different from 72%.

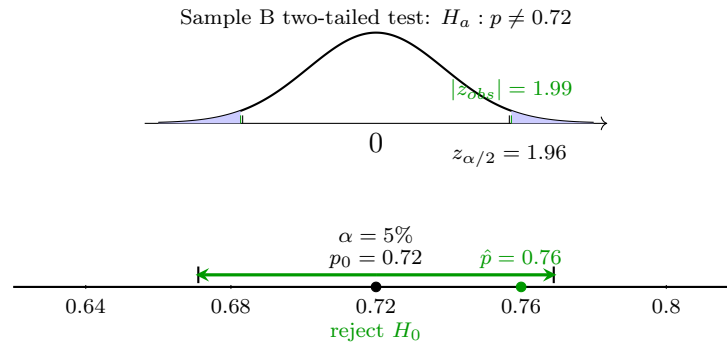


Sample B

- **Conditions:** $np_0 = 500(0.72) = 360 \geq 10$, $n(1 - p_0) = 500(0.28) = 140 \geq 10$.
- **Sample proportion:** $\hat{p} = 380/500 = 0.76$.
- **Test statistic:**

$$z = \frac{0.76 - 0.72}{\sqrt{\frac{0.72(0.28)}{500}}} \approx 1.99.$$

- **p-value:** $2P(Z \geq 1.99) \approx 0.0464$.
- **Decision:** Reject H_0 since $0.0464 < 0.05$.
- **Conclusion:** There is enough evidence that the true on-time submission rate is different from 72%.



Although both samples have the same sample proportion $\hat{p} = 0.76$, the larger sample gives a larger absolute test statistic and a smaller p-value because the standard error is smaller.

5 Platform adoption rate after training (higher-tail test compared across three significance levels)

A district dashboard states that 40% of teachers regularly use a new digital platform. After training, a random sample of 300 teachers shows 138 regular users. At significance levels $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$, test whether the true

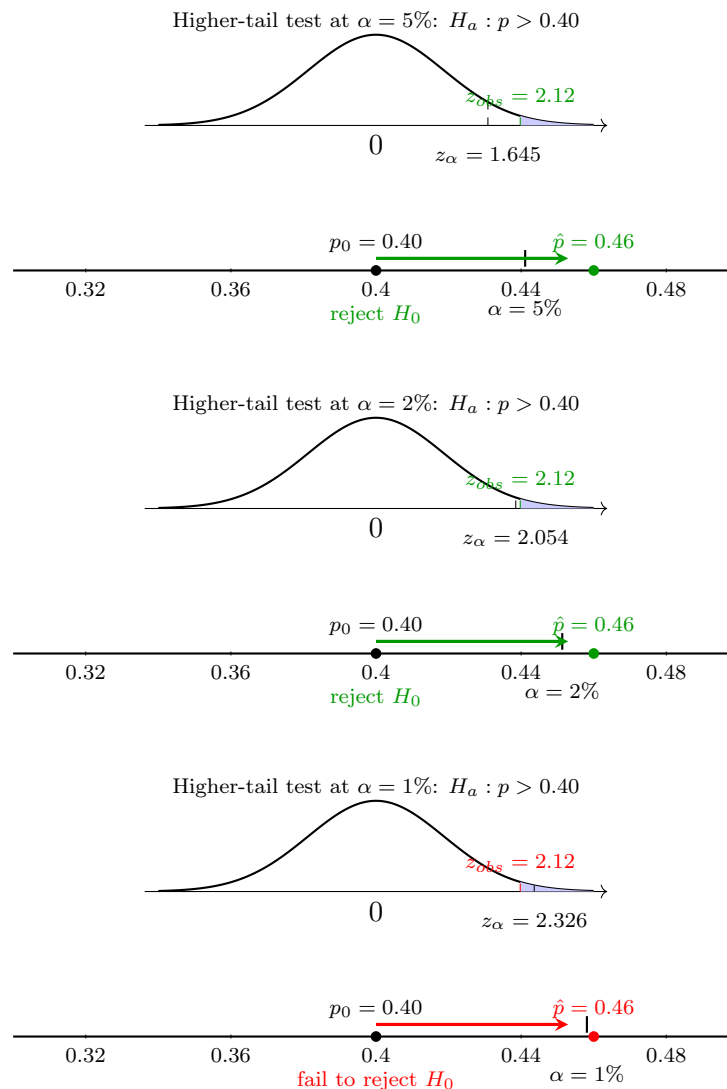
adoption proportion is higher than 40%.

C4

- **Parameter:** p = true regular-use proportion.
- **Hypotheses at $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$:** $H_0: p = 0.40$, $H_a: p > 0.40$.
- **Conditions:** $np_0 = 120 \geq 10$, $n(1 - p_0) = 180 \geq 10$.
- **Sample proportion:** $\hat{p} = 138/300 = 0.46$.
- **Test statistic:**

$$z = \frac{0.46 - 0.40}{\sqrt{\frac{0.40(0.60)}{300}}} \approx 2.12.$$

- **p-value:** $P(Z \geq 2.12) \approx 0.0170$.
- **Decision at $\alpha = 5\%$:** Reject H_0 since $0.0170 < 0.05$.
- **Decision at $\alpha = 2\%$:** Reject H_0 since $0.0170 < 0.02$.
- **Decision at $\alpha = 1\%$:** Fail to reject H_0 since $0.0170 > 0.01$.
- **Conclusion:** The null hypothesis is rejected at the 5% and 2% significance levels, but it is not rejected at the 1% significance level. Therefore, only one of the three significance levels fails to reject H_0 .



6 Platform adoption rate after training (lower-tail test compared across three significance levels)

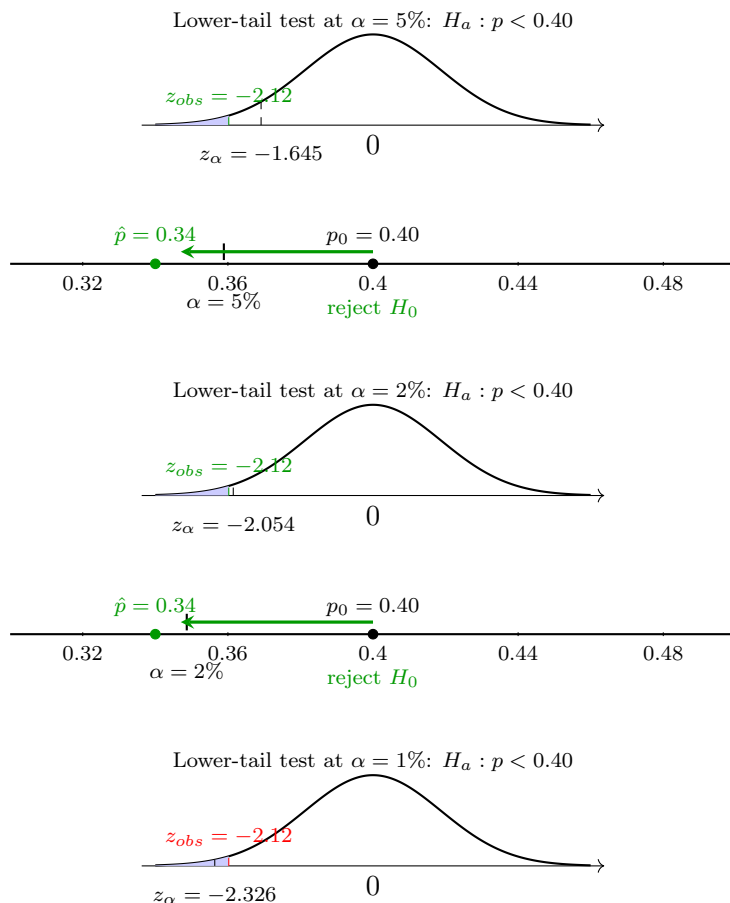
A district dashboard states that 40% of teachers regularly use a new digital platform. After training, a random sample of 300 teachers shows 102 regular users. At significance levels $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$, test whether the true adoption proportion is lower than 40%.

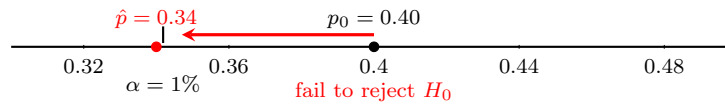
C4

- **Parameter:** p = true regular-use proportion.
- **Hypotheses at $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$:** $H_0: p = 0.40$, $H_a: p < 0.40$.
- **Conditions:** $np_0 = 120 \geq 10$, $n(1 - p_0) = 180 \geq 10$.
- **Sample proportion:** $\hat{p} = 102/300 = 0.34$.
- **Test statistic:**

$$z = \frac{0.34 - 0.40}{\sqrt{\frac{0.40(0.60)}{300}}} \approx -2.12.$$

- **p-value:** $P(Z \leq -2.12) \approx 0.0170$.
- **Decision at $\alpha = 5\%$:** Reject H_0 since $0.0170 < 0.05$.
- **Decision at $\alpha = 2\%$:** Reject H_0 since $0.0170 < 0.02$.
- **Decision at $\alpha = 1\%$:** Fail to reject H_0 since $0.0170 > 0.01$.
- **Conclusion:** The null hypothesis is rejected at the 5% and 2% significance levels, but it is not rejected at the 1% significance level. Therefore, only one of the three significance levels fails to reject H_0 .





7 Platform adoption rate after training (two-tailed test compared across three significance levels)

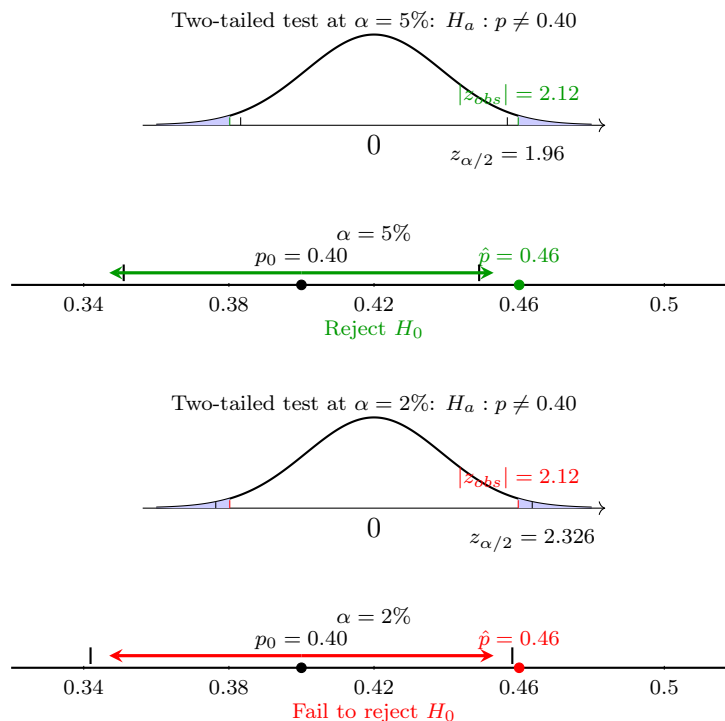
A district dashboard states that 40% of teachers regularly use a new digital platform. After training, a random sample of 300 teachers shows 138 regular users. At significance levels $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$, test whether the true adoption proportion is different from 40%.

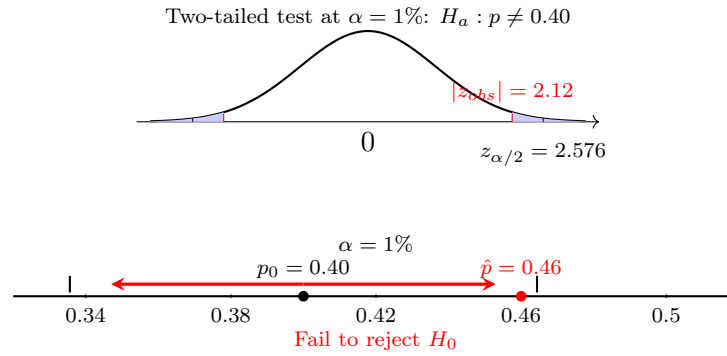
C4

- **Parameter:** p = true regular-use proportion.
- **Hypotheses at $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$:** $H_0: p = 0.40$, $H_a: p \neq 0.40$.
- **Conditions:** $np_0 = 120 \geq 10$, $n(1 - p_0) = 180 \geq 10$.
- **Sample proportion:** $\hat{p} = 138/300 = 0.46$.
- **Test statistic:**

$$z = \frac{0.46 - 0.40}{\sqrt{\frac{0.40(0.60)}{300}}} \approx 2.12.$$

- **p-value:** $2P(Z \geq 2.12) \approx 0.0340$.
- **Decision at $\alpha = 5\%$:** Reject H_0 since $0.0340 < 0.05$.
- **Decision at $\alpha = 2\%$:** Fail to reject H_0 since $0.0340 > 0.02$.
- **Decision at $\alpha = 1\%$:** Fail to reject H_0 since $0.0340 > 0.01$.
- **Conclusion:** The null hypothesis is rejected at the 5% significance level, but it is not rejected at the 2% and 1% significance levels.





8 On-time assignment completion rate (comparison of lower-tail and higher-tail tests in two samples)

A school reports that 72% of assignments are submitted on time. Consider two random samples:

- Sample A has 250 assignments, of which 168 are submitted on time.
- Sample B has 500 assignments, of which 377 are submitted on time.

For learning purposes, apply both one-sided tests to each sample, even though in Sample A the observed proportion is below 72% and in Sample B the observed proportion is above 72%. At $\alpha = 5\%$, test whether the true on-time proportion is lower than 72% and whether it is higher than 72% for each sample.

C4

- **Parameter:** p = true on-time submission proportion.

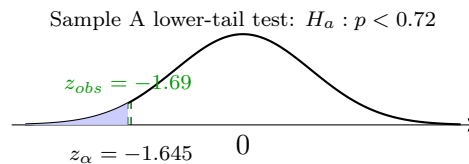
Sample A

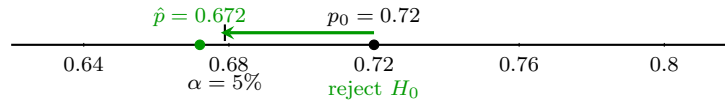
- **Conditions:** $np_0 = 250(0.72) = 180 \geq 10$, $n(1 - p_0) = 250(0.28) = 70 \geq 10$.
- **Sample proportion:** $\hat{p} = 168/250 = 0.672$.
- **Test statistic:**

$$z = \frac{0.672 - 0.72}{\sqrt{\frac{0.72(0.28)}{250}}} \approx -1.69.$$

Test 1: Lower-tail test for Sample A

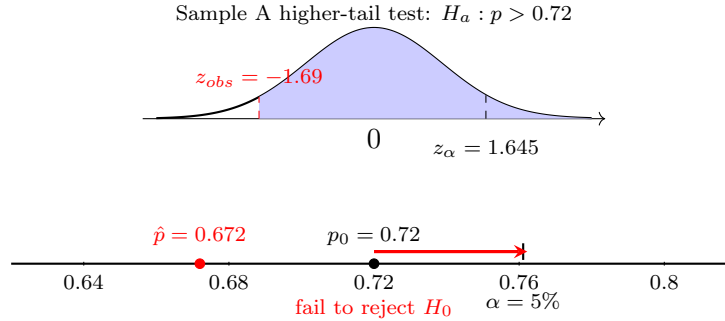
- **Hypotheses at $\alpha = 5\%$:** $H_0 : p = 0.72$, $H_a : p < 0.72$.
- **p-value:** $P(Z \leq -1.69) \approx 0.0455$.
- **Decision:** Reject H_0 since $0.0455 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true on-time submission rate is below 72%.





Test 2: Higher-tail test for Sample A

- **Hypotheses at $\alpha = 5\%$:** $H_0: p = 0.72$, $H_a: p > 0.72$.
- **p-value:** $P(Z \geq -1.69) \approx 0.9545$.
- **Decision:** Fail to reject H_0 since $0.9545 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is above 72%.



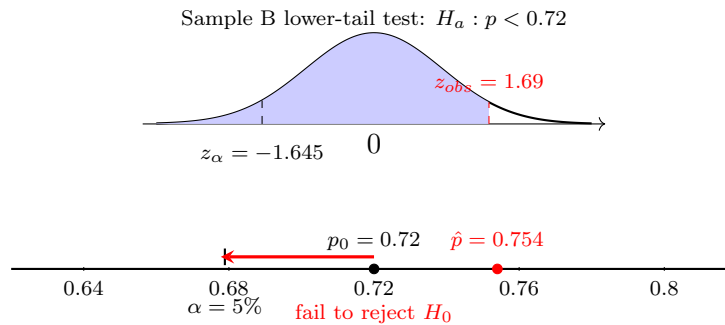
Sample B

- **Conditions:** $np_0 = 500(0.72) = 360 \geq 10$, $n(1 - p_0) = 500(0.28) = 140 \geq 10$.
- **Sample proportion:** $\hat{p} = 377/500 = 0.754$.
- **Test statistic:**

$$z = \frac{0.754 - 0.72}{\sqrt{\frac{0.72(0.28)}{500}}} \approx 1.69.$$

Test 1: Lower-tail test for Sample B

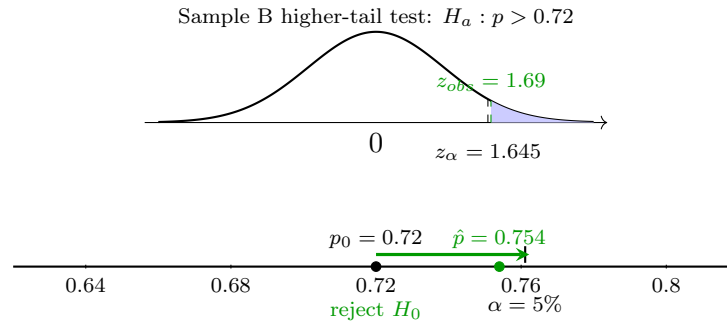
- **Hypotheses at $\alpha = 5\%$:** $H_0: p = 0.72$, $H_a: p < 0.72$.
- **p-value:** $P(Z \leq 1.69) \approx 0.9545$.
- **Decision:** Fail to reject H_0 since $0.9545 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is below 72%.



Test 2: Higher-tail test for Sample B

- **Hypotheses at $\alpha = 5\%$:** $H_0: p = 0.72$, $H_a: p > 0.72$.

- **p-value:** $P(Z \geq 1.69) \approx 0.0455$.
- **Decision:** Reject H_0 since $0.0455 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true on-time submission rate is above 72%.



In Sample A, the observed proportion is below the benchmark, so the lower-tail alternative is the direction supported by the data. In Sample B, the observed proportion is above the benchmark, so the higher-tail alternative is the direction supported by the data.

9 On-time assignment completion rate (comparison of lower-tail and higher-tail tests in two samples, all fail to reject)

A school reports that 72% of assignments are submitted on time. Consider two random samples:

- Sample A has 250 assignments, of which 173 are submitted on time.
- Sample B has 500 assignments, of which 372 are submitted on time.

For learning purposes, apply both one-sided tests to each sample, even though in Sample A the observed proportion is below 72% and in Sample B the observed proportion is above 72%. At $\alpha = 5\%$, test whether the true on-time proportion is lower than 72% and whether it is higher than 72% for each sample.

C4

- **Parameter:** p = true on-time submission proportion.

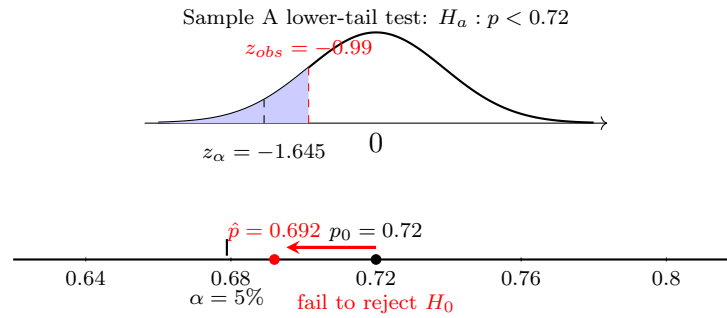
Sample A

- **Conditions:** $np_0 = 250(0.72) = 180 \geq 10$, $n(1 - p_0) = 250(0.28) = 70 \geq 10$.
- **Sample proportion:** $\hat{p} = 173/250 = 0.692$.
- **Test statistic:**

$$z = \frac{0.692 - 0.72}{\sqrt{\frac{0.72(0.28)}{250}}} \approx -0.99.$$

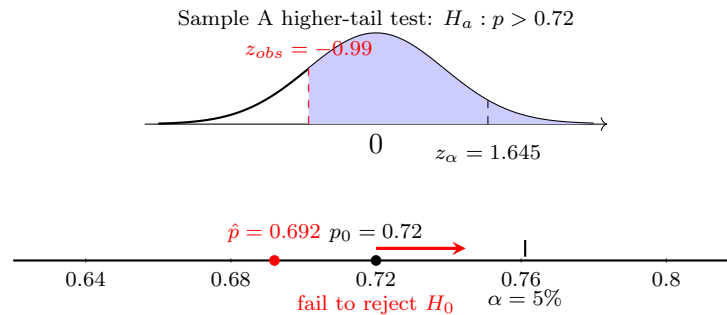
Test 1: Lower-tail test for Sample A

- **Hypotheses at $\alpha = 5\%$:** $H_0 : p = 0.72$, $H_a : p < 0.72$.
- **p-value:** $P(Z \leq -0.99) \approx 0.1611$.
- **Decision:** Fail to reject H_0 since $0.1611 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is below 72%.



Test 2: Higher-tail test for Sample A

- **Hypotheses at $\alpha = 5\%$:** $H_0 : p = 0.72$, $H_a : p > 0.72$.
- **p-value:** $P(Z \geq -0.99) \approx 0.8389$.
- **Decision:** Fail to reject H_0 since $0.8389 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is above 72%.



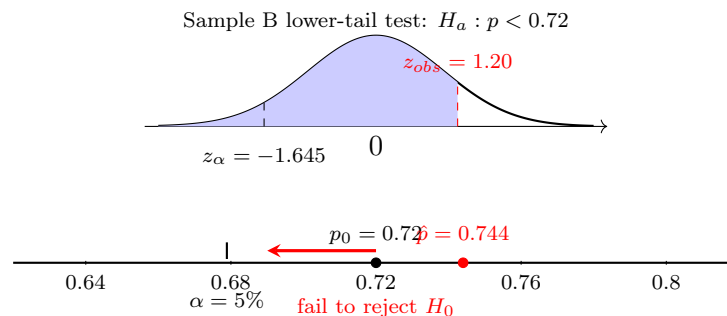
Sample B

- **Conditions:** $np_0 = 500(0.72) = 360 \geq 10$, $n(1 - p_0) = 500(0.28) = 140 \geq 10$.
- **Sample proportion:** $\hat{p} = 372/500 = 0.744$.
- **Test statistic:**

$$z = \frac{0.744 - 0.72}{\sqrt{\frac{0.72(0.28)}{500}}} \approx 1.20.$$

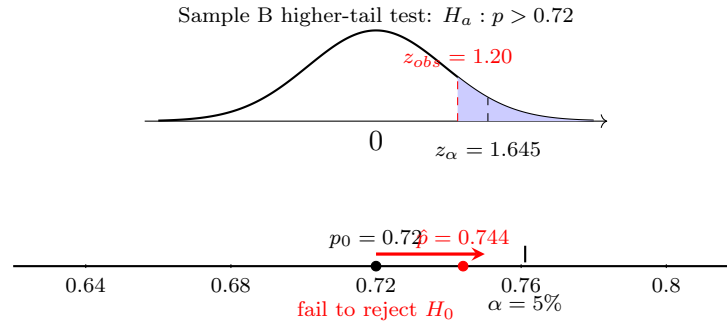
Test 1: Lower-tail test for Sample B

- **Hypotheses at $\alpha = 5\%$:** $H_0 : p = 0.72$, $H_a : p < 0.72$.
- **p-value:** $P(Z \leq 1.20) \approx 0.8849$.
- **Decision:** Fail to reject H_0 since $0.8849 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is below 72%.



Test 2: Higher-tail test for Sample B

- **Hypotheses at $\alpha = 5\%$:** $H_0: p = 0.72$, $H_a: p > 0.72$.
- **p-value:** $P(Z \geq 1.20) \approx 0.1151$.
- **Decision:** Fail to reject H_0 since $0.1151 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission rate is above 72%.



In Sample A, the observed proportion is below the benchmark, so the lower-tail alternative is the direction supported by the data. In Sample B, the observed proportion is above the benchmark, so the higher-tail alternative is the direction supported by the data. However, in both samples the evidence is not strong enough at the 5% significance level, so all tests fail to reject H_0 .

10 Compliance with safety checklist (Comparison of higher-tail and two-tailed I)

A quality team states that 85% of shifts complete a safety checklist. In a random sample of 320 shifts, 288 completed the checklist. At $\alpha = 5\%$, test both whether the true completion proportion is higher than 85% and whether the true completion proportion is different from 85%.

C4

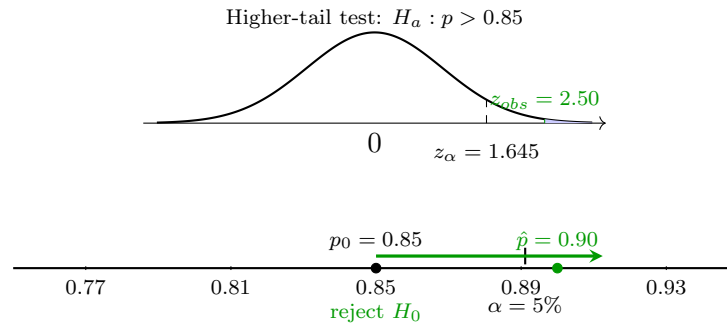
- **Parameter:** p = true checklist-completion proportion.
- **Conditions:** $np_0 = 272 \geq 10$, $n(1 - p_0) = 48 \geq 10$.
- **Sample proportion:** $\hat{p} = 288/320 = 0.90$.

Test 1: Higher-tail test

- **Hypotheses at $\alpha = 5\%$:** $H_0: p = 0.85$, $H_a: p > 0.85$.
- **Test statistic:**

$$z = \frac{0.90 - 0.85}{\sqrt{\frac{0.85(0.15)}{320}}} \approx 2.50.$$

- **p-value:** $P(Z \geq 2.50) \approx 0.0062$.
- **Decision:** Reject H_0 since $0.0062 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true checklist-completion rate is higher than 85%.

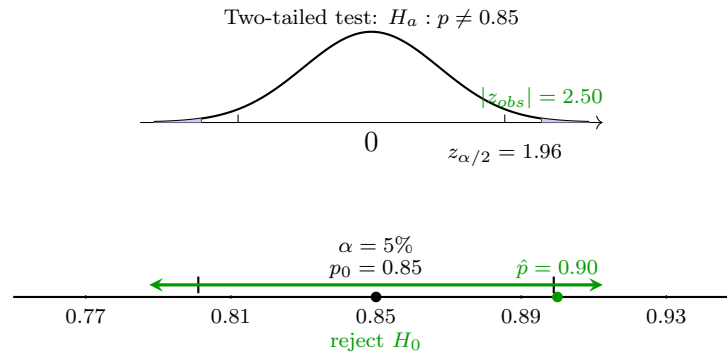


Test 2: Two-tailed test

- **Hypotheses at $\alpha = 5\%$:** $H_0 : p = 0.85$, $H_a : p \neq 0.85$.
- **Test statistic:**

$$z = \frac{0.90 - 0.85}{\sqrt{\frac{0.85(0.15)}{320}}} \approx 2.50.$$

- **p-value:** $2P(Z \geq 2.50) \approx 0.0124$.
- **Decision:** Reject H_0 since $0.0124 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true checklist-completion rate is different from 85%.



11 Compliance with safety checklist (Comparison of higher-tail and two-tailed II)

A quality team states that 85% of shifts complete a safety checklist. In a random sample of 320 shifts, 282 completed the checklist. At $\alpha = 5\%$, test both whether the true completion proportion is higher than 85% and whether the true completion proportion is different from 85%.

C4

- **Parameter:** p = true checklist-completion proportion.
- **Conditions:** $np_0 = 272 \geq 10$, $n(1 - p_0) = 48 \geq 10$.
- **Sample proportion:** $\hat{p} = 282/320 = 0.88125$.

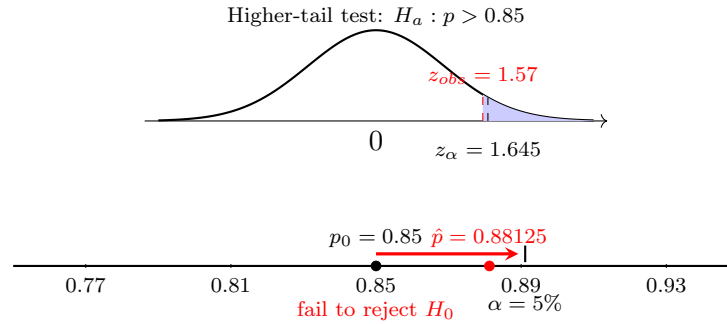
Test 1: Higher-tail test

- **Hypotheses at $\alpha = 5\%$:** $H_0 : p = 0.85$, $H_a : p > 0.85$.

- **Test statistic:**

$$z = \frac{0.88125 - 0.85}{\sqrt{\frac{0.85(0.15)}{320}}} \approx 1.57.$$

- **p-value:** $P(Z \geq 1.57) \approx 0.0582$.
- **Decision:** Fail to reject H_0 since $0.0582 > 0.05$.
- **Conclusion:** There is not enough evidence that the true checklist-completion rate is higher than 85%.

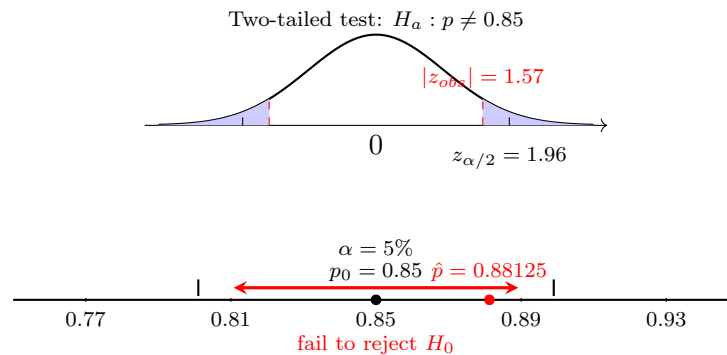


Test 2: Two-tailed test

- **Hypotheses at $\alpha = 5\%$:** $H_0 : p = 0.85$, $H_a : p \neq 0.85$.
- **Test statistic:**

$$z = \frac{0.88125 - 0.85}{\sqrt{\frac{0.85(0.15)}{320}}} \approx 1.57.$$

- **p-value:** $2P(Z \geq 1.57) \approx 0.1164$.
- **Decision:** Fail to reject H_0 since $0.1164 > 0.05$.
- **Conclusion:** There is not enough evidence that the true checklist-completion rate is different from 85%.



12 Compliance with safety checklist (Comparison of higher-tail and two-tailed III)

A quality team states that 85% of shifts complete a safety checklist. In a random sample of 320 shifts, 284 completed the checklist. At $\alpha = 5\%$, test both whether the true completion proportion is higher than 85% and whether the true completion proportion is different from 85%.

C4

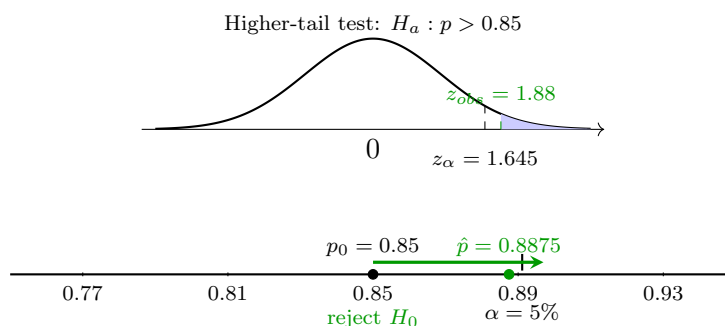
- **Parameter:** p = true checklist-completion proportion.
- **Conditions:** $np_0 = 272 \geq 10$, $n(1 - p_0) = 48 \geq 10$.
- **Sample proportion:** $\hat{p} = 284/320 = 0.8875$.

Test 1: Higher-tail test

- **Hypotheses:** $H_0: p = 0.85$, $H_a: p > 0.85$.
- **Test statistic:**

$$z = \frac{0.8875 - 0.85}{\sqrt{\frac{0.85(0.15)}{320}}} \approx 1.88.$$

- **p-value:** $P(Z \geq 1.88) \approx 0.0301$.
- **Decision:** Reject H_0 since $0.0301 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true checklist-completion rate is higher than 85%.

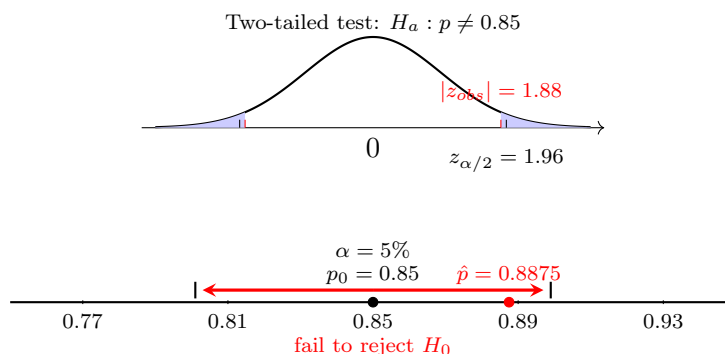


Test 2: Two-tailed test

- **Hypotheses:** $H_0: p = 0.85$, $H_a: p \neq 0.85$.
- **Test statistic:**

$$z = \frac{0.8875 - 0.85}{\sqrt{\frac{0.85(0.15)}{320}}} \approx 1.88.$$

- **p-value:** $2P(Z \geq 1.88) \approx 0.0602$.
- **Decision:** Fail to reject H_0 since $0.0602 > 0.05$.
- **Conclusion:** There is not enough evidence that the true checklist-completion rate is different from 85%.



13 On-time assignment completion rate (higher-tail test with fixed sample proportion and variable sample size)

A school reports that 72% of assignments are submitted on time. Suppose that for samples of different sizes, the observed sample proportion is always

$$\hat{p} = 0.76.$$

At significance levels $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$, test whether the true on-time proportion is higher than 72%, and describe how the value of the test statistic changes as the sample size n increases.

C4

- **Parameter:** p = true on-time submission proportion.
- **Hypotheses at $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$:** $H_0: p = 0.72$, $H_a: p > 0.72$.
- **Condition:** For a one-sample proportion test, check

$$np_0 = n(0.72) \geq 10 \quad \text{and} \quad n(1 - p_0) = n(0.28) \geq 10.$$

Since

$$n \geq \frac{10}{0.28} \approx 35.72,$$

the conditions are satisfied for $n \geq 36$.

- **Sample proportion:** $\hat{p} = 0.76$.
- **Test statistic as a function of n :**

$$z(n) = \frac{0.76 - 0.72}{\sqrt{\frac{0.72(0.28)}{n}}}.$$

Simplifying,

$$z(n) = \frac{0.04}{\sqrt{\frac{0.2016}{n}}} = \frac{0.04\sqrt{n}}{\sqrt{0.2016}} \approx 0.0891\sqrt{n}.$$

- **p-value as a function of n :**

$$p\text{-value}(n) = P(Z \geq z(n)).$$

Since this is a higher-tail test,

$$p\text{-value}(n) = 1 - \Phi(z(n)),$$

where Φ is the standard normal cumulative distribution function.

- **Interpretation of the formulas:** Since $z(n) \approx 0.0891\sqrt{n}$, the test statistic increases as n increases when the sample proportion stays fixed at 0.76. Therefore, the p-value decreases as n increases.
- **Critical values:** For a higher-tail test,

$$z_{10\%} = 1.282, \quad z_{5\%} = 1.645, \quad z_{1\%} = 2.326.$$

- **When does rejection begin?** Reject H_0 when $z(n) > z_\alpha$.

At $\alpha = 10\%$,

$$0.0891\sqrt{n} > 1.282$$

so

$$\sqrt{n} > \frac{1.282}{0.0891} \approx 14.39, \quad n > 207.0.$$

So rejection begins for

$$n \geq 208.$$

At $\alpha = 5\%$,

$$0.0891\sqrt{n} > 1.645$$

so

$$\sqrt{n} > \frac{1.645}{0.0891} \approx 18.46, \quad n > 340.9.$$

So rejection begins for

$$n \geq 341.$$

At $\alpha = 1\%$,

$$0.0891\sqrt{n} > 2.326$$

so

$$\sqrt{n} > \frac{2.326}{0.0891} \approx 26.11, \quad n > 681.5.$$

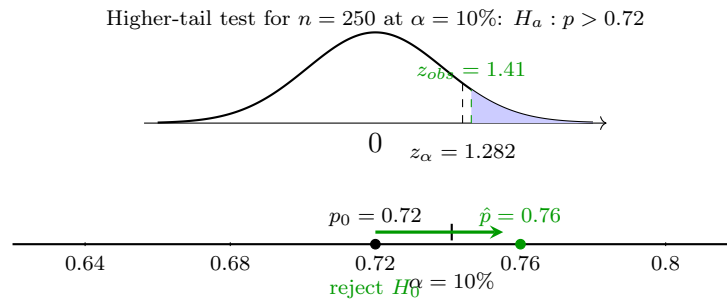
So rejection begins for

$$n \geq 682.$$

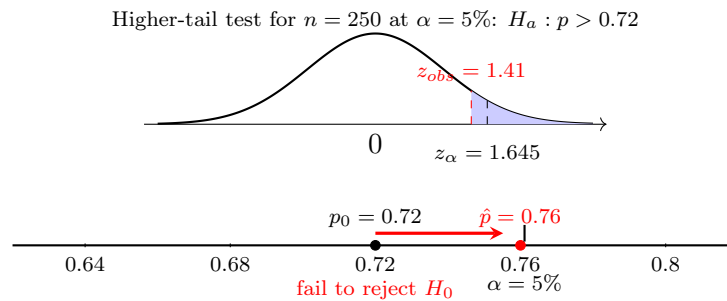
• **Sample 1:** $n = 250$.

$$z(250) \approx 0.0891\sqrt{250} \approx 1.41, \quad p\text{-value}(250) \approx 0.0795.$$

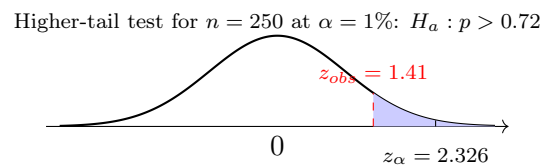
At $\alpha = 10\%$, reject H_0 since $0.0795 < 0.10$.

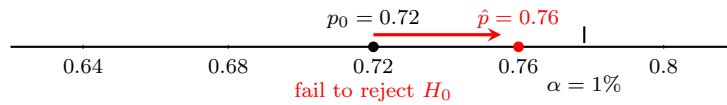


At $\alpha = 5\%$, fail to reject H_0 since $0.0795 > 0.05$.



At $\alpha = 1\%$, fail to reject H_0 since $0.0795 > 0.01$.

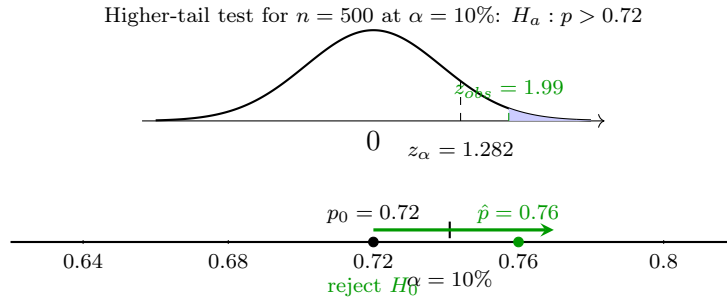




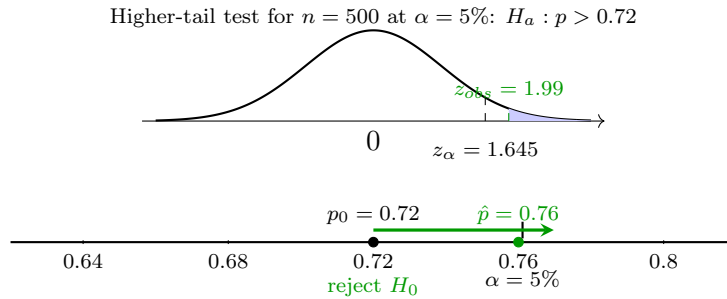
- **Sample 2:** $n = 500$.

$$z(500) \approx 0.0891\sqrt{500} \approx 1.99, \quad p\text{-value}(500) \approx 0.0232.$$

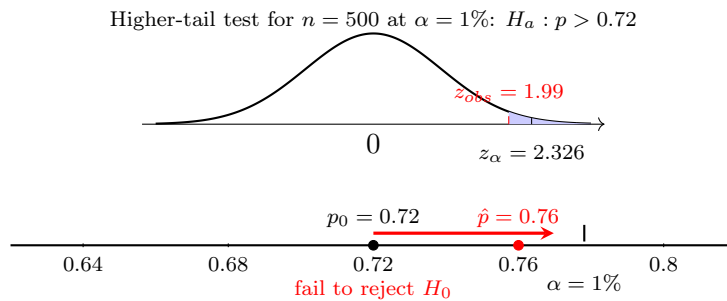
At $\alpha = 10\%$, reject H_0 since $0.0232 < 0.10$.



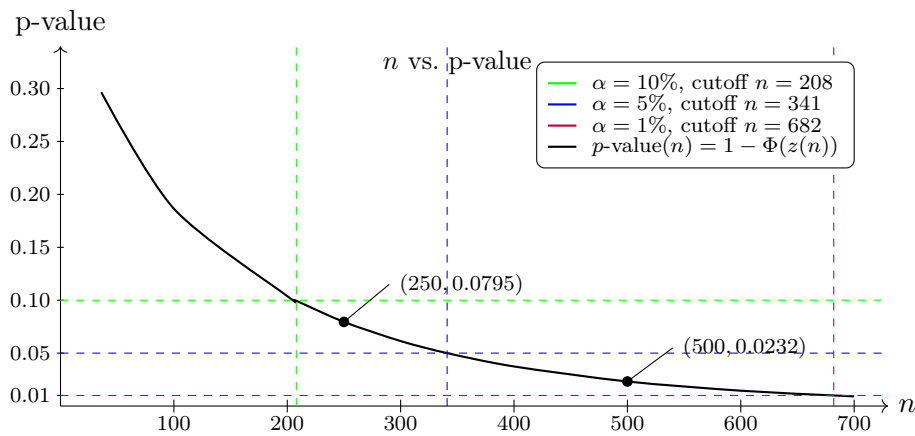
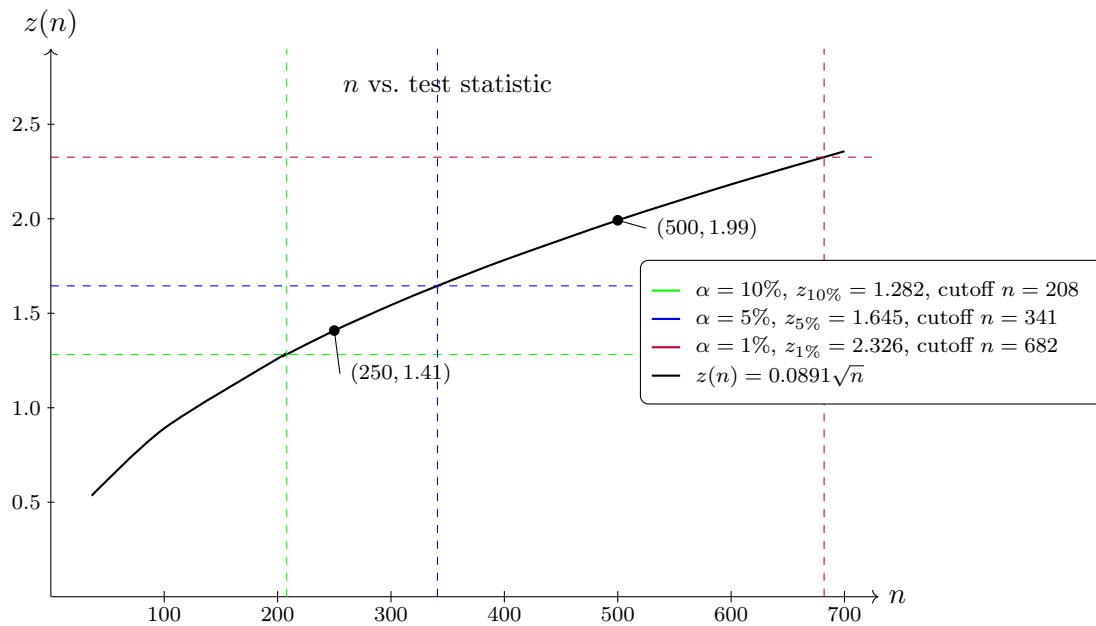
At $\alpha = 5\%$, reject H_0 since $0.0232 < 0.05$.



At $\alpha = 1\%$, fail to reject H_0 since $0.0232 > 0.01$.



- **Conclusion:** If the sample proportion remains fixed at 0.76, then increasing the sample size increases the test statistic and decreases the p-value. As a result, rejection of H_0 first occurs at the 10% level, then at the 5% level, and finally at the 1% level as n becomes large enough.



14 Workshop attendance rate (higher-tail test with fixed sample size and variable sample proportion)

A school reports that 72% of students attend the weekly academic support workshop. Suppose that the sample size is fixed at

$$n = 400,$$

and that the observed sample proportion $\hat{p}$ varies from one sample to another. At significance levels $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$, test whether the true attendance proportion is higher than 72%, and describe how the value of the test statistic and the p-value change as the observed sample proportion changes.

C4

- **Parameter:** p = true workshop-attendance proportion.
- **Hypotheses at $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$:**

$$H_0: p = 0.72, \quad H_a: p > 0.72.$$

- **Conditions:** For a one-sample proportion test, check

$$np_0 = 400(0.72) = 288 \geq 10 \quad \text{and} \quad n(1 - p_0) = 400(0.28) = 112 \geq 10.$$

So the normal conditions are satisfied.

- **Test statistic as a function of $\hat{p}$:**

$$z(\hat{p}) = \frac{\hat{p} - 0.72}{\sqrt{\frac{0.72(0.28)}{400}}}.$$

Since

$$\sqrt{\frac{0.72(0.28)}{400}} = \sqrt{0.000504} \approx 0.02245,$$

we obtain

$$z(\hat{p}) \approx \frac{\hat{p} - 0.72}{0.02245} \approx 44.54(\hat{p} - 0.72).$$

- **p-value as a function of $\hat{p}$:** Since this is a higher-tail test,

$$p\text{-value}(\hat{p}) = P(Z \geq z(\hat{p})) = 1 - \Phi(z(\hat{p})).$$

- **Interpretation of the formulas:** As $\hat{p}$ increases, the quantity $\hat{p} - 0.72$ increases, so the test statistic $z(\hat{p})$ increases. Since the p-value is

$$1 - \Phi(z(\hat{p})),$$

the p-value decreases as $\hat{p}$ increases.

- **Critical values:**

$$z_{10\%} = 1.282, \quad z_{5\%} = 1.645, \quad z_{1\%} = 2.326.$$

- **Rejection thresholds in terms of $\hat{p}$:** Reject H_0 when

$$z(\hat{p}) > z_\alpha.$$

At $\alpha = 10\%$,

$$44.54(\hat{p} - 0.72) > 1.282,$$

so

$$\hat{p} - 0.72 > \frac{1.282}{44.54} \approx 0.0288,$$

and therefore

$$\hat{p} > 0.7488.$$

At $\alpha = 5\%$,

$$44.54(\hat{p} - 0.72) > 1.645,$$

so

$$\hat{p} - 0.72 > \frac{1.645}{44.54} \approx 0.0369,$$

and therefore

$$\hat{p} > 0.7569.$$

At $\alpha = 1\%$,

$$44.54(\hat{p} - 0.72) > 2.326,$$

so

$$\hat{p} - 0.72 > \frac{2.326}{44.54} \approx 0.0522,$$

and therefore

$$\hat{p} > 0.7722.$$

- **Example 1:** Suppose the sample gives

$$\hat{p} = 0.7525.$$

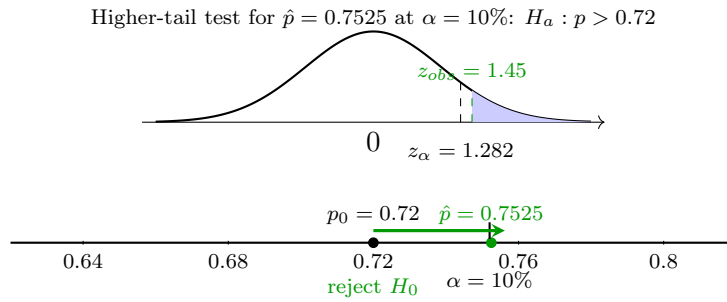
Then

$$z(0.7525) = \frac{0.7525 - 0.72}{\sqrt{\frac{0.72(0.28)}{400}}} \approx 1.45.$$

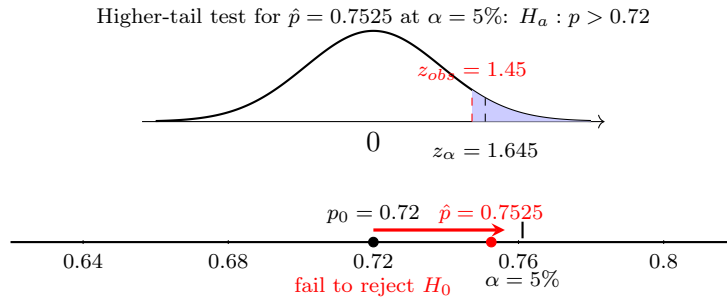
The corresponding p-value is

$$p\text{-value}(0.7525) = P(Z \geq 1.45) \approx 0.0739.$$

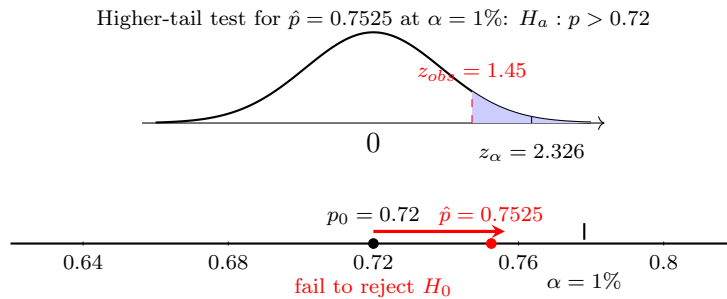
At $\alpha = 10\%$, reject H_0 since $0.0739 < 0.10$.



At $\alpha = 5\%$, fail to reject H_0 since $0.0739 > 0.05$.



At $\alpha = 1\%$, fail to reject H_0 since $0.0739 > 0.01$.



- **Example 2:** Suppose the sample gives

$$\hat{p} = 0.7650.$$

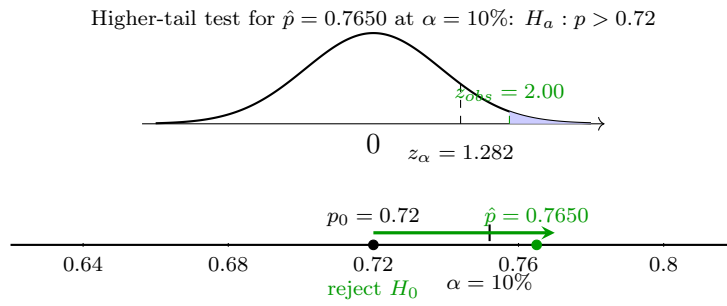
Then

$$z(0.7650) = \frac{0.7650 - 0.72}{\sqrt{\frac{0.72(0.28)}{400}}} \approx 2.00.$$

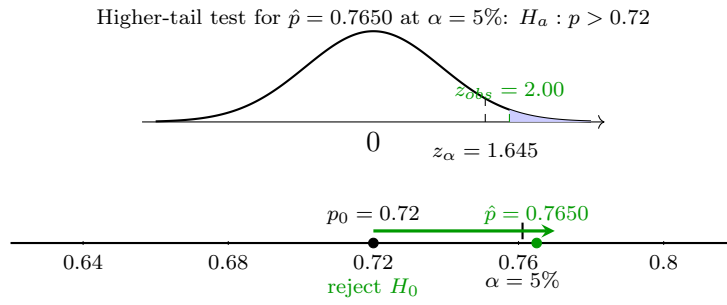
The corresponding p-value is

$$p\text{-value}(0.7650) = P(Z \geq 2.00) \approx 0.0225.$$

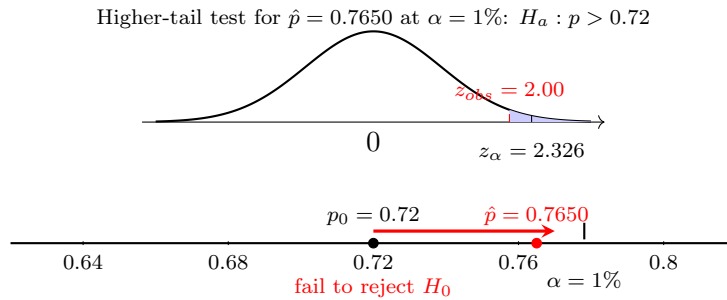
At $\alpha = 10\%$, reject H_0 since $0.0225 < 0.10$.



At $\alpha = 5\%$, reject H_0 since $0.0225 < 0.05$.



At $\alpha = 1\%$, fail to reject H_0 since $0.0225 > 0.01$.



- **Conclusion:** With the sample size fixed at $n = 400$, increasing the observed sample proportion $\hat{p}$ increases the test statistic and decreases the p-value. Therefore, rejection begins once $\hat{p}$ becomes large enough. It starts at the 10% level, then at the 5% level, and finally at the 1% level as $\hat{p}$ increases.

